

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B4. NUCLEAR AND PARTICLE PHYSICS

TRINITY TERM 2022

Wednesday, 8th June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) Draw a representative leading order Feynman diagram showing how K^0 mesons can transform into $\bar{K}^0$ mesons, thus allowing their states to mix. [2]

(b) Two different orthogonal mixtures of neutral kaon states can be defined as $|+\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle + |\bar{K}^0\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|K^0\rangle - |\bar{K}^0\rangle)$. The combined application of the symmetry operators C and P interchanges the identities of particle and antiparticle, as well as changing the parity. Given that the state $|+\rangle$ is even under CP, and $|-\rangle$ is odd, deduce the effect of C. Both of these states can decay into multiple pions. For each CP state, how many pions result from such a decay? Draw a representative Feynman diagram for each decay. To a good approximation, these different CP states correspond to a shorter lived K_S^0 and a longer lived K_L^0 . Explain why these states have different lifetimes and indicate to which CP states they are related. [7]

(c) Determine the threshold energy in terms of the particle masses to produce neutral kaons with a beam of protons interacting on a fixed target via the reaction:

$$p + p \rightarrow p + \pi^+ + \Lambda^0 + K^0.$$

Draw a quark flow diagram for this process. Neutral kaons produced in this manner emerge as K_S^0 and K_L^0 , which differ in mass by just 3.49×10^{-12} MeV/c². These are then allowed to travel some distance in vacuum so that the K_S^0 component decays away. The resulting beam of surviving K_L^0 mesons then travels through another thin block of material. Discuss and explain the composition of the beam in terms of the kaon CP states after it has passed through the block. [6]

(d) In natural units ($\hbar=c=1$), the amplitude for the wavefunction of the decaying K_S^0 state can be written as:

$$A_S(t) = A_S(0) \exp(-im_S t) \exp(-t/(2\tau_S)).$$

A similar expression can also be written for K_L^0 . Give a qualitative justification of this form. Derive expressions for the wavefunction amplitudes of K_0 and $\bar{K}^0$ in terms of $A_S(t)$ and $A_L(t)$. [3]

(e) A beam of pure K_0 mesons are created at time $t = 0$. Show that expressions for the observed intensity of K_0 and $\bar{K}^0$ as a function of time have the following form:

$$I = Ae^{-t/\tau_S} + Be^{-t/\tau_L} + C \cos(\phi).$$

Determine the factors A , B , C and the phase angle ϕ for K_0 and $\bar{K}^0$, respectively. Experimentally, $\tau_S = 8.958 \times 10^{-11}$ s and $\tau_L = 5.18 \times 10^{-8}$ s. On the same plot, sketch the intensities of K_0 and $\bar{K}^0$ components as a function of time in units of τ_S over the range $0 < t/\tau_S < 10$. [7]

2. (a) Define each of the terms in Fermi's Golden Rule #2: $\Gamma = \frac{2\pi}{\hbar} |M|^2 \rho$. [2]

(b) A sub-dominant decay mode of the π^+ meson is $\pi^+ \rightarrow \mu^+ \nu_\mu \gamma$. Draw a representative Feynman diagram for this process and use phase space considerations to derive an approximate expression for how the branching ratio for this decay depends on the pion and muon masses, stating all assumptions. [5]

(c) What additional factors need to be considered in scaling from the lifetime of the muon ($\tau_\mu \sim 2.2 \mu\text{s}$) to that of the τ lepton? From these, estimate the lifetime of the τ lepton. [4]

(d) What is the relationship between Fermi's Golden Rule #2 and Feynman diagrams? Illustrate the main components of a Feynman diagram and describe what each represents. [3]

(e) Draw the leading order Feynman diagrams for the processes of Compton scattering, bremsstrahlung and pair production. [3]

(f) Draw the leading order Feynman diagrams for the following elastic scattering processes and comment on expectations for the relative cross sections:

- (i) $e^- \nu_e \rightarrow e^- \nu_e$;
- (ii) $e^- \nu_\mu \rightarrow e^- \nu_\mu$;
- (iii) $e^- \bar{\nu}_e \rightarrow e^- \bar{\nu}_e$;
- (iv) $e^+ \bar{\nu}_e \rightarrow e^+ \bar{\nu}_e$. [5]

(g) Consider the following process: $\mu^- \rightarrow e^- e^- e^+ \nu_\mu \bar{\nu}_e$.

Is this allowed within the Standard Model? If not, explain why not. If so, draw a representative Feynman diagram for this. [3]

3. (a) In terms of the parent (P) and daughter (D) neutral atomic rest masses and other relevant quantities, determine the Q values for β^- decay, β^+ decay and electron capture. [3]

(b) Determine the kinetic energy of an α particle emitted from a nucleus of atomic number A in terms of the Q value for the reaction. [3]

(c) The dependence of the mean lifetime, τ , of an α -unstable nucleus on the value of Z and velocity, v , of the emitted α particle can be approximated by:

$$\tau \propto \exp(Ze^2/(\epsilon_0 \hbar v)).$$

Describe and comment on the basic assumptions made in the derivation of this expression. [3]

(d) Radioactive material a (decay constant λ_a) decays into a material b (decay constant λ_b), which is also radioactive. The initial amounts of material are N_{a0} and N_{b0} , respectively. The amount of material b remaining after a time t is:

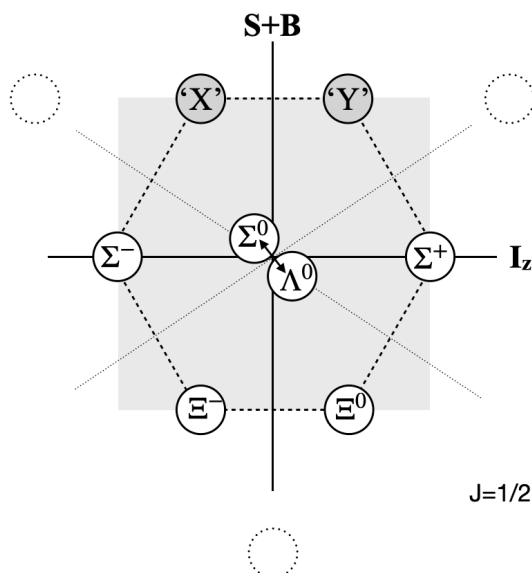
$$N_b = \alpha e^{-\lambda_a t} + \beta e^{-\lambda_b t}.$$

(i) Determine the factors α and β . [8]

(ii) If $N_b = 0$ at $t = 0$, determine the time at which N_b is maximum. [4]

(iii) If material b decays into a stable substance c , determine how the amount, N_c , varies with time, again assuming $N_b = 0$ at $t = 0$. [4]

4. In the early days of the Standard Model, a plot of the combination of baryon number (B) and “Strangeness” (S, which has a negative sign convention), versus the z-component of Isospin (I_z) revealed the following pattern for the then known $J=1/2$ baryons:



(a) What is the meaning of B, S and I_z in the context of the quark model? Show explicitly how the observed pattern is reproduced with a 3-quark model and identify the particles 'X' and 'Y' in the diagram above. [6]

(b) Why is there only 1 state at each of the edge positions given the different possible spin combinations that can achieve $J = 1/2$, and why are there no states at the corner positions indicated by dotted circles? [3]

(c) Why are there 2 states in the middle and what are these? [2]

(d) What measured properties of the particles would indicate that they are likely composed of 3 quarks and do not involve combinations of quarks and antiquarks? [2]

(e) What linear combination of (S+B) and I_z can be related to electrical charge of the particle? Derive the required charges of u,d and s quarks. [2]

(f) The Σ^0 decays at rest to Λ^0 and one other particle. What must this particle be and why? Determine the energy of this particle in terms of the Σ^0 and Λ^0 rest masses. Why are the decays $\Sigma^0 \rightarrow n + \pi^0$ and $\Sigma^0 \rightarrow p + \pi^-$ not observed? [4]

(g) Use the quark model to construct a similar plot as above for the $J = 0$ meson nonet. Identify those particles that are pions or kaons and discuss the nature of the 3 states in the middle of the diagram. [6]